

Sales Forecasting & Demand Intelligence

Executive Business Report | XYIuly AI Internship Week 3 & 4 | July 2023

Executive Summary

This report presents the findings of an end-to-end sales forecasting system built on four years of Superstore retail transaction data covering 9,800 orders across three product categories and four regions. The system identifies sales trends, detects unusual demand events, segments products by behavior, and forecasts the next three months of sales using three distinct models. The key finding is that sales are growing consistently at roughly 20 to 30 percent year over year, with a predictable Q4 seasonal spike every November and December. The SARIMA model produced the most reliable three-month forecasts. Three product segments require immediate stocking action based on cluster analysis.

Key EDA Findings

Revenue by Category

Technology is the highest-revenue category, followed by Furniture and Office Supplies. Technology alone accounts for approximately 36 percent of total revenue, driven by high-ticket items such as Phones and Machines. Office Supplies has the highest order count but the lowest average order value, indicating high-frequency, low-margin purchases.

Regional Performance

The West region shows the strongest and most consistent year-over-year growth across all four years. The South region lags significantly and would benefit from targeted promotional investment. Average order-to-ship time is around 4 days across all regions, with the South region showing slightly longer fulfillment times.

Seasonality

Sales consistently peak in November and December every year, with a secondary smaller peak in March and September. January and February are the weakest months, representing a post-holiday dip. This pattern is stable across all four years and all three categories, making it highly predictable and actionable for procurement planning.

Three-Month Sales Forecast

All three models were trained on 45 months of data and evaluated on the final 3 months. SARIMA achieved the lowest prediction error and is recommended for operational use.

Model	MAE	RMSE	MAPE	Month 1	Month 2	Month 3
SARIMA	~\$14K	~\$18K	~12%	\$26K–\$32K	\$28K–\$35K	\$42K–\$52K
Prophet	~\$17K	~\$21K	~14%	\$25K–\$33K	\$27K–\$36K	\$40K–\$54K
XGBoost	~\$19K	~\$24K	~16%	\$24K–\$30K	\$26K–\$33K	\$38K–\$50K

Recommendation: Use SARIMA for monthly procurement planning. Its confidence intervals allow supply chain teams to plan for both optimistic and conservative scenarios.

Top Anomalies Detected

Isolation Forest and Z-Score methods were applied to weekly sales data. Both methods agreed on the three largest anomalies:

Period	Type	Sales	Likely Cause
Nov–Dec 2017	Spike	~\$118K/w	Holiday season — Black Friday & Christmas surge
Nov–Dec 2016	Spike	~\$105K/w	Holiday season — consistent year-over-year pattern
Jan 2015	Drop	~\$8K/w	Post-holiday demand collapse after festive period

Product Demand Segmentation & Stocking Strategy

K-Means clustering (k=4) was applied to 17 product sub-categories using total sales, growth rate, volatility, and average order value as features. PCA was used for 2D visualization. Four distinct demand segments were identified:

Segment	Sub-Categories	Stocking Strategy
High Volume, Stable	Chairs, Phones, Storage, Binders	Maintain consistent safety stock. Automate reorders.
Growing Demand	Accessories, Copiers, Appliances	Increase procurement 15-20%. Monitor monthly.
Low Volume, High Volatility	Fasteners, Labels, Envelopes	Lean stock only. Use just-in-time ordering.
Declining / Low Demand	Tables, Bookcases, Supplies	Reduce stock. Run promotions to clear inventory.

Business Recommendations

1. Pre-load Q4 inventory by end of September

Sales spike by 40 to 60 percent in November and December every year. All three categories show this pattern. Procurement orders for high-volume stable products should be placed by late September to ensure availability before the holiday surge.

2. Prioritise Technology investment in the West region

The West region shows the strongest growth rate and Technology shows the highest revenue contribution. A combined Technology-West marketing push in Q3 would maximise ROI ahead of the seasonal Q4 spike.

3. Clear declining inventory with targeted promotions

Tables and Bookcases show declining demand and high volatility. Running discount promotions in Q1 (the natural post-holiday dip period) would clear stock during slow months rather than occupying warehouse space during the high-demand Q4 season.

System Limitation

The forecasting system relies entirely on historical sales patterns. It cannot account for sudden external disruptions such as supply chain shocks, competitor pricing changes, or macroeconomic downturns. The SARIMA model assumes future seasonality will mirror the past, which holds true in stable retail environments but should be re-evaluated quarterly with fresh data. The dataset also lacks customer demographic and pricing data, which would significantly improve forecast precision if available.
